

Setting up your computer

NSDF-Data student guide 1 — Linux edition

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Tip

Who this is for. You are starting from little or no computing experience. You will paste some commands into a terminal and read the answers. You do **not** need to know any programming. Work through the steps in order. Each step ends with a **Checkpoint**: if it does not match, stop there.

Before you start

- **Time:** about one hour. Most of it is waiting for downloads and installers.
- **You need:** an internet connection, the password for your computer, and about 3 GB of free disk space.
- **Rule of thumb:** type or paste commands **exactly** as written. Capitals and spaces matter.
- **When something goes wrong:** do not guess. Copy the message, note the step you were on, and ask the project lead. See *If something goes wrong* at the end.

Words you will meet

Word	What it means
Terminal	A window where you type commands and the computer answers in text.
Command	One line you type in the terminal, followed by Enter (or Return).
Folder (or directory)	A place on your computer that holds files and other folders.
Path	The address of a folder, such as C:\Users\YourName\Research.
Git	A program that downloads and keeps track of versions of code.
Repository (repo)	A project folder that Git tracks. Here, NSDF-Data.
Python	The programming language the analysis is written in. You will not need to write Python to finish these guides.

Word	What it means
Conda / Miniforge	A tool that installs Python and packages in separate, named <i>environments</i> .
Environment	A sealed set of programs and versions, so this project's software does not clash with others.
Package	A piece of ready-made software (for example <i>numpy</i>).
Jupyter notebook	A document that mixes text, code and plots, run one piece (<i>cell</i>) at a time.
Branch	A separate line of work in Git, so your changes do not touch the main code.

Step 1. Open a terminal

Press **Ctrl + Alt + T** (on Ubuntu), or search your applications for **Terminal**. A window with a text prompt opens. This is the terminal for everything in this guide.

Paste with **Ctrl + Shift + V** (plain Ctrl+V does not work in most terminals).

Checkpoint

You see a line of text ending in a symbol such as `>`, `$` or `%`. Type `echo hello` and press Enter: the word `hello` should appear under it.

Step 2. Install Git

This guide uses Ubuntu or Debian commands. If you use another distribution, use its package manager instead. Type:

```
sudo apt install git
```

It asks for your password. Nothing appears while you type it; that is normal. Press **Return**, then type `y` and press **Return** if it asks to continue.

Checkpoint

In the terminal, type `git --version` and press Enter. You should see a line starting with `git version`.

Step 3. Install Miniforge

Miniforge installs Python and the other packages the analysis needs. Its download file is **Miniforge3-Linux-x86_64.sh** (most PCs).

Check your computer type with this command, which prints one word:

```
uname -m
```

`x86_64` means the file below. `aarch64` means an ARM computer: use **Miniforge3-Linux-aarch64.sh** instead.

Open github.com/conda-forge/miniforge in your web browser and download the file into your

Downloads folder. Then install it with this command (use the file name you downloaded):

```
bash ~/Downloads/Miniforge3-Linux-x86_64.sh
```

Press **Return** to read the licence, type yes, accept the default location, and when it asks about running `conda init`, type yes. Then **close the terminal and open a new one**.

Checkpoint

Open a **new** terminal and type `conda --version`, then press Enter. You should see `conda` followed by a version number.

Careful

Use `sudo` only for the one command that installs Git. Other commands in this guide must not use it.

Step 4. Download the NSDF-Data code

Your project will live in a folder called `Research`. Create it, go into it, and download the code. Type each line and press Enter:

```
mkdir -p ~/Research
cd ~/Research
git clone https://github.com/villano-lab/NSDF-Data.git
cd ~/Research/NSDF-Data
```

Downloading with `git clone` does not need a GitHub account. It makes a full copy of the project in a folder named `NSDF-Data`.

Checkpoint

Type `ls` and press Enter. You should see `NSDF-Data` in the list.

Step 5. Create your analysis environment

An environment is a sealed box of programs with fixed versions. We make one for this project, called `darkmatter_cli_env`. Type these lines one at a time, pressing Enter after each. The third line takes several minutes and prints a lot of text: wait for the prompt to come back.

```
conda create -n darkmatter_cli_env python=3.10 -y
conda activate darkmatter_cli_env
python -m pip install nsdf-dark-matter==0.3.0 nsdf-dark-matter-cli==0.5.0
python -m pip install numpy==2.2.6 matplotlib==3.10.7 h5py pytest ipykernel
python -m pip install jupyterlab
```

Careful

Do **not** use the file `environment.yml` from the project. It was made on a Mac for Mac only, and it will fail on other computers. The three lines above install the same versions.

Tip

Every time you open a new terminal, run `conda activate darkmatter_cli_env` first. When it is active, the start of each line shows `(darkmatter_cli_env)`.

Checkpoint

The last line of the output is a message, not an error, and the `(darkmatter_cli_env)` label is at the start of your prompt.

Step 6. Check that everything works

First, the tests. They use made-up data, so they need no download. Type each line and press Enter:

```
conda activate darkmatter_cli_env
cd ~/Research/NSDF-Data/python
python -m pytest -q
```

Checkpoint

The last line says `52 passed` (it can be a larger number later, which is also fine). If it says failed, stop and ask the project lead, and do not change any files.

Then, the data tool:

```
nsdf-cli version
```

Checkpoint

It prints `NSDF Dark Matter CLI: 0.5.0` (or a later version).

Finally, the notebook program:

```
jupyter lab --version
```

Checkpoint

It prints a version number, such as `4.4.0`. If it says `jupyter-lab` is not found, run `python -m pip install jupyterlab` and try again.

Step 7. Let Jupyter notebooks use your environment

Jupyter is the notebook program you will use. Register your environment with it once:

```
python -m ipykernel install --user --name darkmatter_cli_env
```

Checkpoint

It prints a line that starts with `Installed kernelspec darkmatter_cli_env`.

Step 8. Get permission to save your work

You will not change the main project. You will save your work on your own **branch**, and for that the lead must give you access:

1. Make a free account at **github.com** if you do not have one.
2. Send your GitHub username to the project lead. They will add you to the project as a collaborator.
3. Once you have accepted the invitation, set your name and email for Git. Use your own details:

```
git config --global user.name "Your Name"
git config --global user.email "you@example.com"
```

4. Create your personal branch. Use your own name, without spaces, in place of yourname:

```
cd ~/Research/NSDF-Data
git checkout -b student-yourname
```

Checkpoint

The terminal shows Switched to a new branch 'student-yourname'.

Rules for keeping your work safe

- **Never work on master.** That is the shared main version. Work only on your student- branch.
- **Never commit data.** The raw files in the idx folder are large and not part of the project. Download only from your home folder, never from inside NSDF-Data.
- **Do not run 07221203_2025_dump1_noise.ipynb** unless the project lead says it is fine. It overwrites a shared archive file.
- **Save often**, and commit your work when a piece is finished (see the first analysis guide).

You are done when

- ☐ Your terminal opens and `git --version` prints a version.
- ☐ `conda --version` prints a version, in a new terminal.
- ☐ The NSDF-Data folder is in `~/Research`.
- ☐ `python -m pytest -q` in the python folder ends with 52 passed.
- ☐ `nsdf-cli` version prints a version.
- ☐ `jupyter lab --version` prints a version.
- ☐ Your Jupyter kernel `darkmatter_cli_env` is installed.
- ☐ You have a student- branch (after access is granted).

Next: *First analysis* (the guide for Linux).

If something goes wrong

1. Read the **last line** of the message. It usually says what is wrong.
2. Close the terminal, open a new one, and run `conda activate darkmatter_cli_env` again. Many problems are just the wrong environment.
3. Copy the command you ran and the message it printed, and send them to the project lead. A screenshot is also fine.

Where to get help

Anthony Villano, project lead: anthony.villano@ucdenver.edu

When you write, include the step number, the command you typed, and the last few lines the computer printed. A screenshot helps. Nobody will be annoyed by a question about a step that did not work.

Session Info

Guide version 3, 6 October 2026, Linux edition. Written for beginners; please tell the project lead where a step was unclear, so the guide can be fixed.

Key links

- Notes site: <https://villano-lab.github.io/NSDF-Data/>
- Repository: <https://github.com/villano-lab/NSDF-Data>
- Miniforge downloads: <https://github.com/conda-forge/miniforge>
- Git downloads: <https://git-scm.com>